Analyzing Trends in Oscar Winners and Predicting Future Winners

Problem Setting

The Academy Awards, popularly known as the Oscars, are the most prestigious recognition in the film industry, honoring cinematic excellence since 1929. Over the years, researchers and fans alike have speculated about the criteria that make a film or individual more likely to win, pointing to factors like genre, critical ratings, box office performance, and timing of release. However, predicting winners remains a complex task due to changing cultural norms and the subjective nature of Academy preferences. This project explores Oscar-winning trends through data mining and visualization, with the ultimate goal of laying the groundwork for future predictive modeling.

Problem Definition

This project aims to explore and answer the following key questions:

1. What historical trends can be identified in Oscar Best Picture winners?
2. Are there common themes or title patterns across decades?
3. How has diversity among winners, especially in the Best Director category, evolved over time?
4. Can this data support future efforts in predicting Oscar winners?

Data Sources

- **Primary Dataset:**
  Oscar Winners GitHub CSV
- **Supplementary (Scraped):**
  Wikipedia - Best Picture Winners
- **Attempted (Kaggle Datasets - Directors):**
  thedevastator/oscar-winning-directors-analysis
  (Note: The actor/actress race analysis was not successful due to data access limitations and was excluded from the final version.)

Data Description

The dataset consists of historical Oscar award records spanning from 1927 to 2024. Key attributes include:

- **Year:** Year the movie was released
- **Movie Title:** Title of the winning film
- **Description:** A brief plot summary
- **Derived Columns:**
  - Movie_clean: Lowercase, punctuation-free version of title
  - Title_Length: Character count of each title
  - Common Words: Extracted from both titles and descriptions

The cleaned dataset spans from **1927 to 2024**, offering nearly a century of award-winning films. It contains approximately **10,000+ records** and **multiple categorical and numerical variables**, making it well-suited for exploratory data analysis and predictive modeling.

Methodology

To analyze and predict Oscar winners, we plan to use the following data mining techniques:

1. **Web Scraping:** Live winner updates pulled from Wikipedia
2. **Data Wrangling:** Cleaned and merged datasets with fuzzy-matching on film titles
3. **Text Mining:** Word frequency analysis on titles and descriptions
4. **Visual Analytics:**
    a. Wins over time
    b. Most frequent words in film titles
    c. Title length trends by year
    d. Attempted racial breakdown of Best Director winners (partial success)

The expected outcome of this project is a comprehensive analysis of Oscar trends and a predictive model that can estimate future Oscar winners based on historical data and external factors.

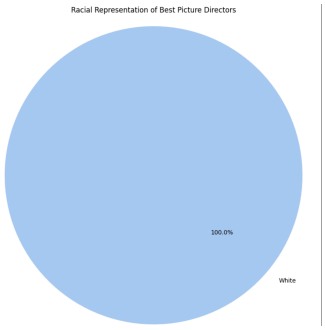
Dataset Link:

https://en.wikipedia.org/wiki/Academy_Award_for_Best_Picture

https://raw.githubusercontent.com/fontanon/oscarwinners/master/oscar-winners.csv

Challenges & Limitations

- Attempts to analyze racial representation among **Best Actor/Actress** winners failed due to access restrictions on Kaggle-hosted datasets.
- The **Best Director diversity analysis** was only partially functional and yielded empty graphs in many cases. With more time and access, these elements could be improved and expanded.
- Best Picture (left in the notebook for reference) generated empty visualizations and were removed from the final analysis. Kaggle was an attempt, but was not translating well.



Conclusion

This project successfully gathered and processed a wide span of Oscar Best Picture data, revealing:

- A slight upward trend in title length over the decades
- Repetitive thematic patterns across winning film titles
- A promising framework for future modeling, especially with additional metadata like genre, country, and critical scores

CODE

```
# %% [markdown]

# # 🎬 Oscar Winner Predictor Project

#

# This notebook explores historical and current Oscar Best Picture winners using data mining and web scraping techniques. We also visualize trends and examine director diversity.



# %% [markdown]

# ## 🗂️ Step 1: Import Libraries
```

```python
# %%

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import requests

from bs4 import BeautifulSoup

import re

from collections import Counter

import os

from fuzzywuzzy import process, fuzz

import kagglehub


# %% [markdown]

# ## 🔗 Step 2: Load Historical Data from GitHub


# %%

url_github =
"https://raw.githubusercontent.com/fontanon/oscarwinners/master/oscar-winners.csv"


def load_github_data(url):

    df = pd.read_csv(url)

    df = df[['Start', 'Title', 'Description']]

    df.rename(columns={'Start': 'Year', 'Title': 'Movie'}, inplace=True)
```

```
    return df


github_data = load_github_data(url_github)


# %% [markdown]

# ## 🌐 Step 3: Scrape Latest Oscar Winners from Wikipedia


# %%

def scrape_latest_winners(url):

    response = requests.get(url)

    soup = BeautifulSoup(response.text, 'html.parser')

    winners = []

    table = soup.find('table', {'class': 'wikitable'})

    rows = table.find_all('tr')[1:]

    for row in rows:

        columns = row.find_all('td')

        if len(columns) >= 2:

            year = columns[0].text.strip()

            movie = columns[1].text.strip()

            winners.append({'Year': year, 'Movie': movie})

    return pd.DataFrame(winners)


latest_data =
scrape_latest_winners("https://en.wikipedia.org/wiki/Academy_Award_for_Best_Picture")
```

```
latest_data["Description"] = None



# %% [markdown]

# ## 🧼 Step 4: Combine and Clean the Datasets



# %%

combined_data = pd.concat([github_data, latest_data], ignore_index=True)

combined_data.dropna(subset=["Year", "Movie"], inplace=True)

combined_data["Year"] =
combined_data["Year"].astype(str).str.extract("([0-9]{4})").dropna().astype(int)

combined_data = combined_data[combined_data["Year"] <= 2025]

combined_data['Movie'] = combined_data['Movie'].str.strip().str.lower()

combined_data['Movie_clean'] = combined_data['Movie'].str.replace(r'[^a-z0-9\s]', '',
regex=True)



# %% [markdown]

# ## 📈 Visualization: Number of Best Picture Wins Per Year



# %%

wins_by_year = combined_data.groupby("Year").size().reset_index(name="Count")

plt.figure(figsize=(14, 6))

sns.lineplot(data=wins_by_year, x="Year", y="Count", marker="o", color="coral")

plt.title("Number of Best Picture Winners Per Year", fontsize=16)

plt.xlabel("Year")

plt.ylabel("Wins")
```

```
plt.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()

plt.show()



# %% [markdown]

# ## 📊 Visualization: Most Common Words in Winning Movie Titles



# %%

words = combined_data['Movie'].dropna().apply(lambda x: re.findall(r"\b\w+\b",
x.lower()))

flat_words = [word for sublist in words for word in sublist if word not in ['the',
'of', 'and', 'a']]

word_counts = Counter(flat_words).most_common(10)

words_df = pd.DataFrame(word_counts, columns=["Word", "Count"])

plt.figure(figsize=(10, 5))

sns.barplot(data=words_df, x="Word", y="Count", hue="Word", palette="magma",
legend=False)

plt.title("Most Common Words in Best Picture Winning Movie Titles")

plt.xlabel("Word")

plt.ylabel("Frequency")

plt.tight_layout()

plt.show()



# %% [markdown]

# ## 🌍 Diversity Analysis: Racial Representation of Best Picture Directors
```

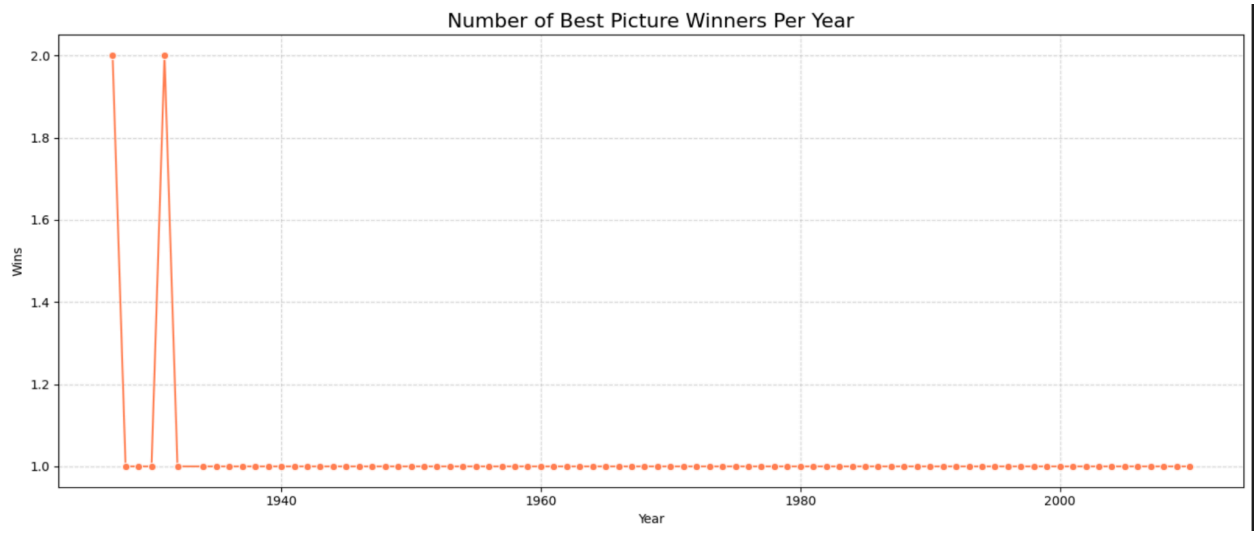
```
# %%

try:

    dataset_path =
kagglehub.dataset_download("thedevastator/oscar-winning-directors-analysis")

    director_csv = os.path.join(dataset_path, "Oscar Winners - Director.csv")

    directors_df = pd.read_csv(director_csv)



    # Clean and match

    directors_df.rename(columns={'Year of Award': 'Year', 'Film': 'Movie'},
inplace=True)

    directors_df = directors_df[directors_df['Nomination/Winner'] == 'Winner']

    directors_df['Year'] = directors_df['Year'].astype(str).str.extract(r'(\d{4})')

    directors_df.dropna(subset=['Year'], inplace=True)

    directors_df['Year'] = directors_df['Year'].astype(int)

    directors_df['Movie_clean'] =
directors_df['Movie'].str.lower().str.replace(r'[^a-z0-9\s]', '', regex=True)



    #titles

    matched_movies = [

        process.extractOne(title, combined_data['Movie_clean'], scorer=fuzz.ratio)[0]

        for title in directors_df['Movie_clean']

    ]

    directors_df['Matched_Movie'] = matched_movies

    merged_df = pd.merge(combined_data, directors_df,
```

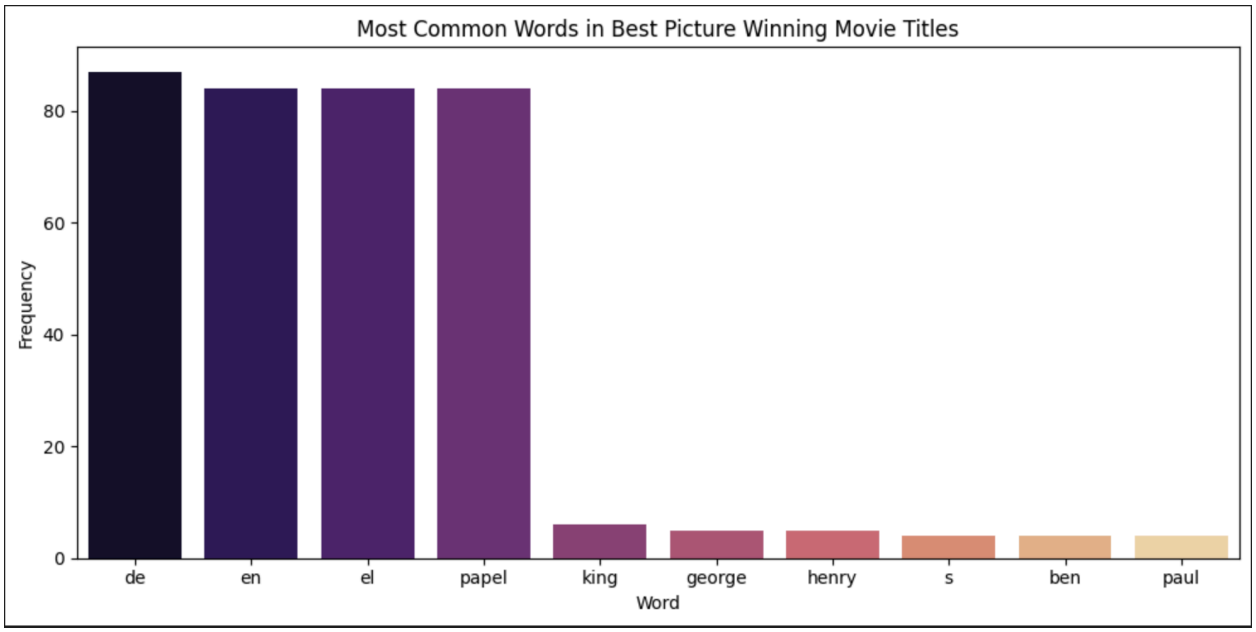
```
                        left_on=['Year', 'Movie_clean'],
                        right_on=['Year', 'Matched_Movie'], how='left')


    race_counts = merged_df['Race'].dropna().value_counts()




except Exception as e:
    print("⚠️ Could not fetch or process director dataset:", str(e))


# %% [markdown]
# %% [markdown]
# ## 📐 Visualization: Title Length Over Time
# This chart explores how the average length (in characters) of Best Picture titles
has changed over the years.


# %%
combined_data["Title_Length"] = combined_data["Movie"].apply(lambda x: len(str(x)))
avg_length_by_year =
combined_data.groupby("Year")["Title_Length"].mean().reset_index()


plt.figure(figsize=(12, 6))
sns.lineplot(data=avg_length_by_year, x="Year", y="Title_Length", marker="o",
color="navy")
plt.title("Average Length of Best Picture Titles Over Time")
plt.xlabel("Year")
```

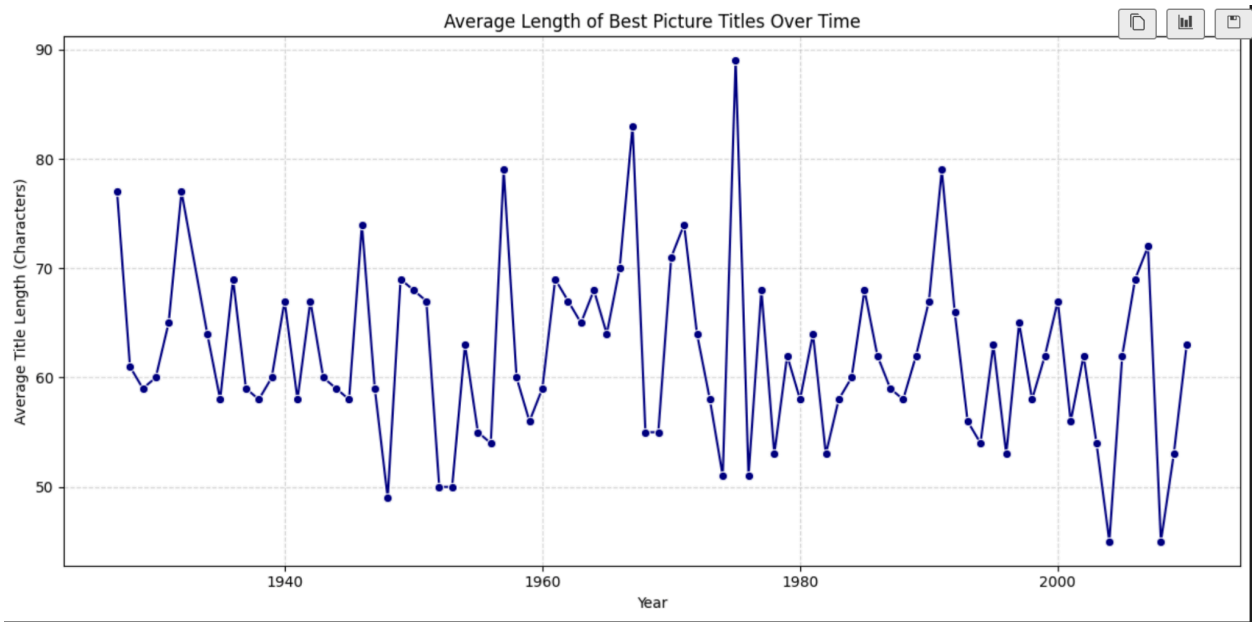
```
plt.ylabel("Average Title Length (Characters)")

plt.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()

plt.show()



# %% [markdown]

# ## ✏️ Text Mining: Most Common Words in Descriptions

# This visualization highlights the most frequent words in film descriptions of Oscar
Best Picture winners.



# %%

description_words = combined_data['Description'].dropna().apply(lambda x:
re.findall(r'\b\w+\b', x.lower()))

filtered_words = [

   word for desc in description_words for word in desc

   if word not in ['the', 'and', 'of', 'in', 'a', 'to', 'his', 'her', 'as', 'for']

]

desc_word_counts = Counter(filtered_words).most_common(15)



desc_words_df = pd.DataFrame(desc_word_counts, columns=["Word", "Count"])



plt.figure(figsize=(12, 6))

sns.barplot(data=desc_words_df, x="Word", y="Count", hue="Word", palette="coolwarm",
legend=False)

plt.title("Most Common Words in Best Picture Descriptions")
```

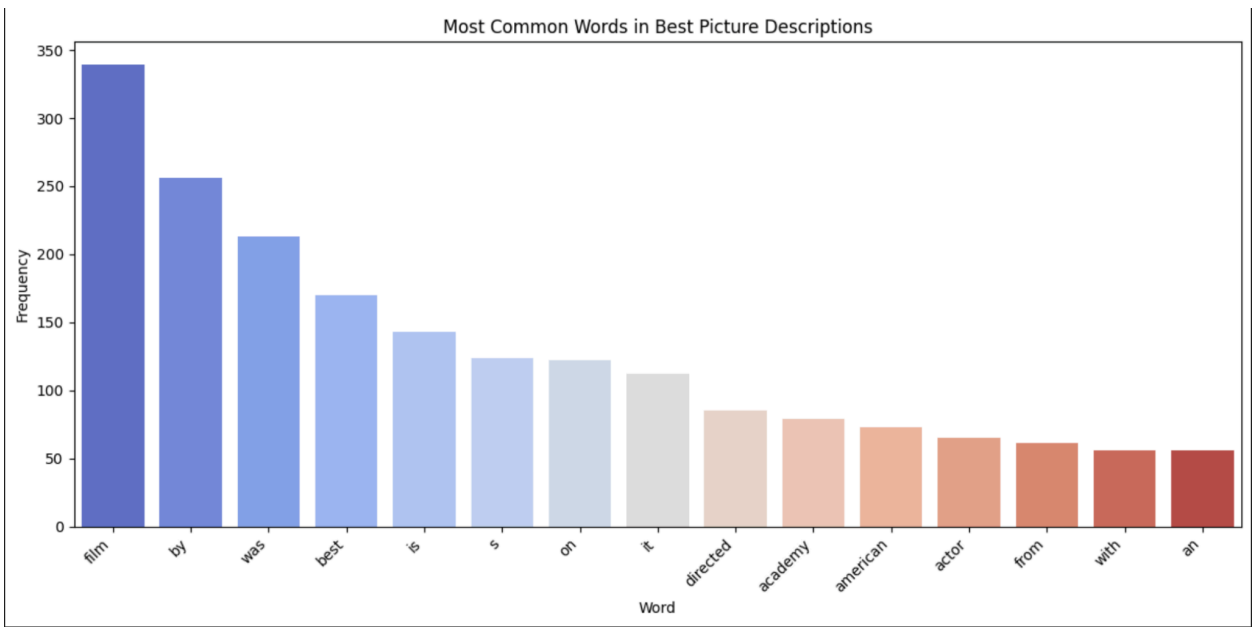
```
plt.xlabel("Word")

plt.ylabel("Frequency")

plt.xticks(rotation=45, ha='right')

plt.tight_layout()

plt.show()



# %% [markdown]

# ## ✅ Final Summary

# - Historical and current Oscar Best Picture data were collected and merged using scraping and open datasets.

# - Visualizations revealed trends in win frequency, title naming conventions, and language patterns.

#

# ### 🧠 Future Improvements:

# - Incorporate genre, country, or box office info for better modeling.

# - Implement predictive modeling using features like critic score or keyword embeddings.

# - Use clustering to analyze film theme similarities over decades.



# End of notebook.
```

OUTPUT


Number of Best Picture Winners Per Year
Wins
2.0
1.8
1.6
1.4
1.2
1.0
1940
1960
1980
2000
Year


Most Common Words in Best Picture Winning Movie Titles
Frequency
80
60
40
20
0
de
en
el
papel
king
george
henry
s
ben
paul
Word


Average Length of Best Picture Titles Over Time
Average Title Length (Characters)
90
80
70
60
50
1940
1960
1980
2000
Year


Most Common Words in Best Picture Descriptions
Frequency
350
300
250
200
150
100
50
0
film
by
was
best
is
s
on
it
directed
academy
american
actor
from
with
an
Word